

2015 GCSE Syllabus Worksheets

Foundation Worksheets

Grades 1 to 5

Grade 1

Grade 2

Grade 3

Grade 4

Grade 5

Part 2

Instructions for use

Clicking on something like this **Grade 4** will take you to the Grade 4 clip numbers and titles.



Clicking on something like this Compound Units. 142 will take you to Worksheet 142.



Clicking on something like this

©MathsWatch	Clip 142
Grade 4 questions	

 will take you back to the contents page.



Grade 2 Foundation and Higher

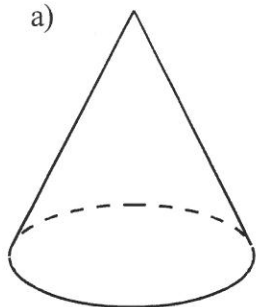
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1) Draw a sketch of each of the following solids:

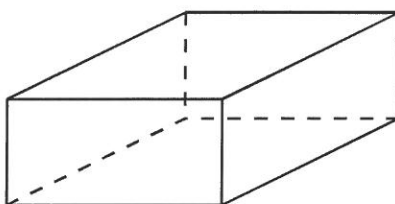
- a) A cube.
- b) A cylinder.

2) Write down the mathematical name of each of these 3-D shapes.

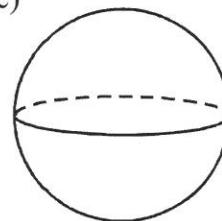
a)



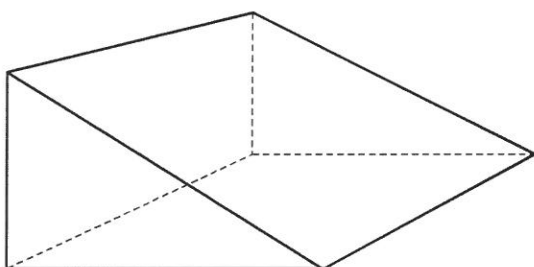
b)



c)



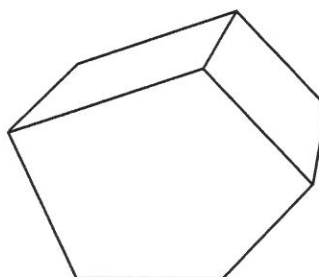
3) Look at this solid.



- a) What is its name?
- b) How many vertices does it have?
- c) How many edges are there?
- d) How many faces does it have?

4) This is a picture of a pentagonal prism.

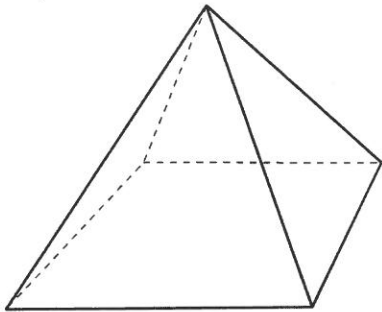
- a) How many faces does it have?
- b) How many edges does it have?
- c) How many vertices does it have?



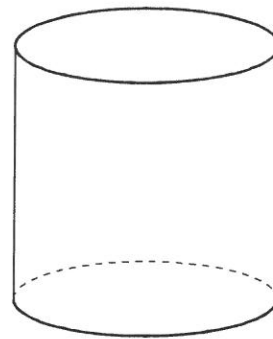
Nets

1) Sketch nets of these solids.

a)



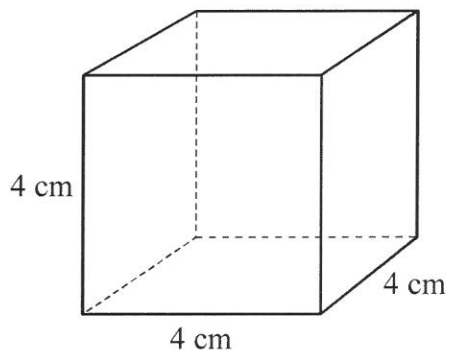
b)



2) On squared paper draw accurate nets of these solids.

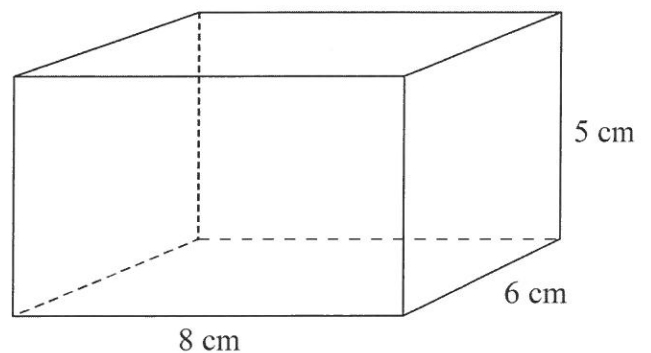
a)

Cube

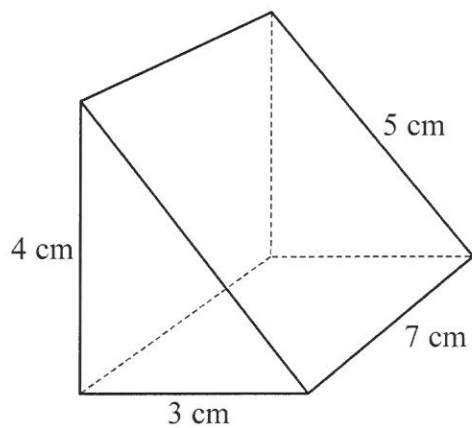


b)

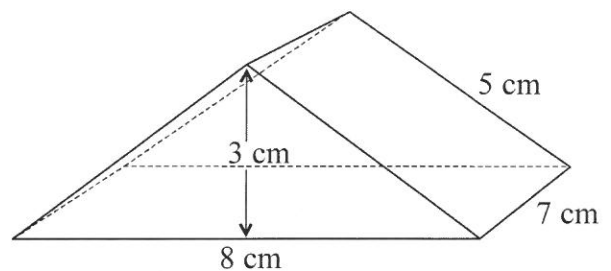
Cuboid



c) **Right-angled triangular prism**

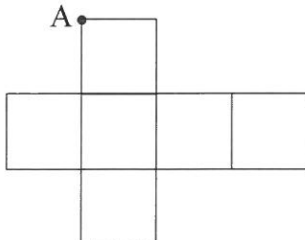


d) **Triangular prism**

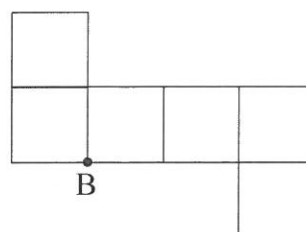


3) The two nets, below, are folded to make cubes.
Two other vertices will meet at the the dot, A. Mark them with As.
One other vertex will meet the dot B. Mark it with B.

a)

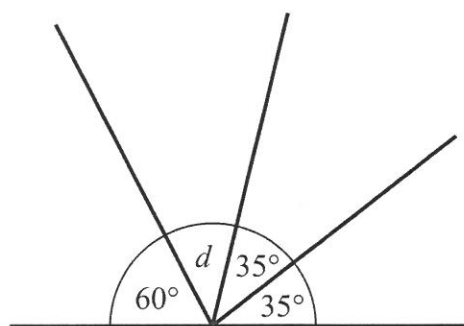
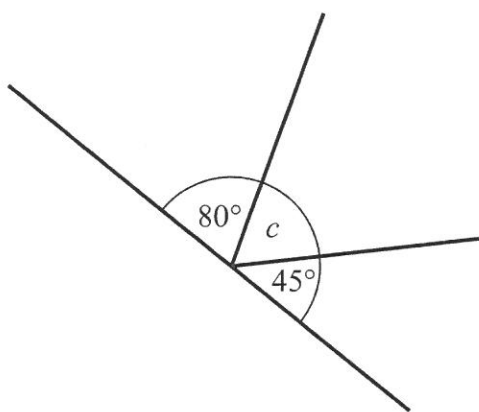
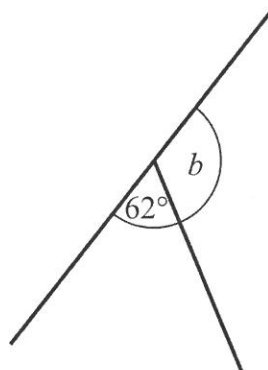
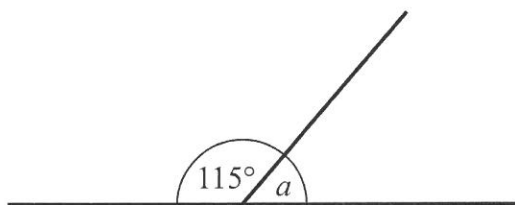


b)

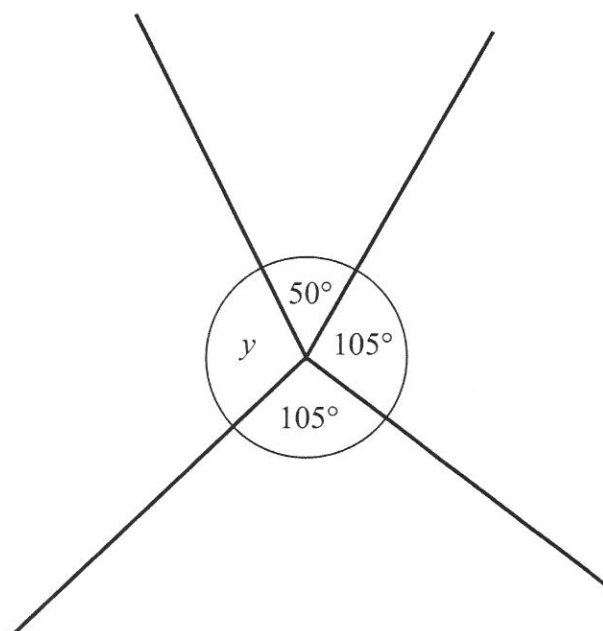
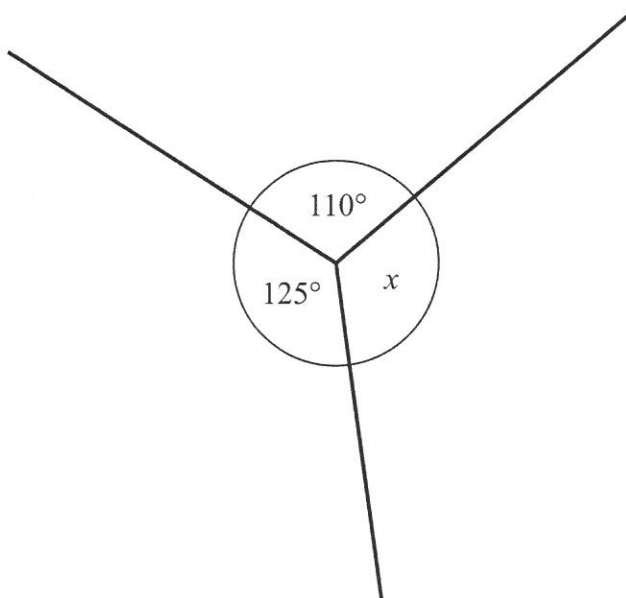


Angles on a Line and at a Point

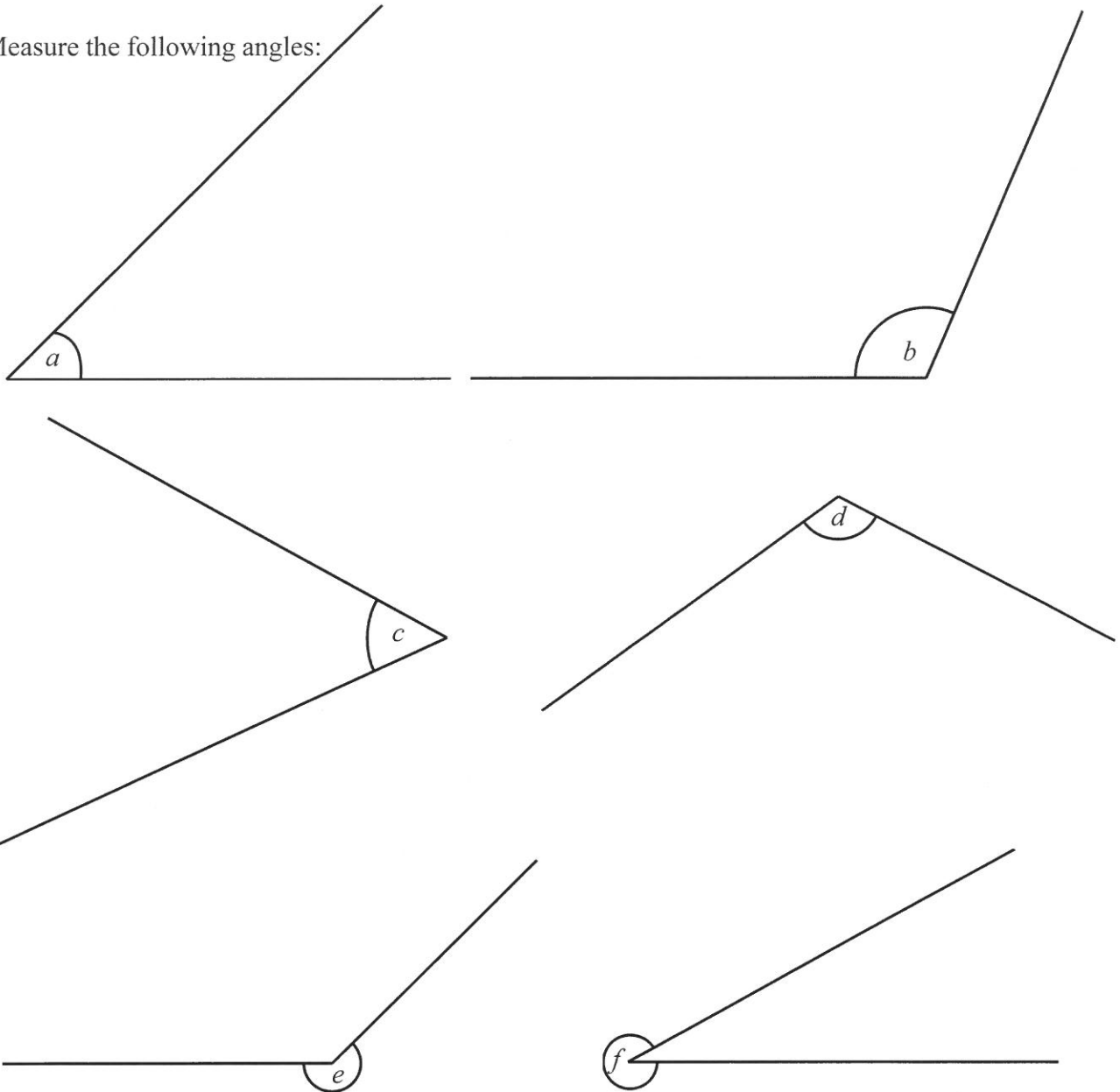
- 1) Work out the values of the unknown angles.



- 2) Work out the values of the unknown angles.



1) Measure the following angles:



2) Draw the following angles:

a) Angle $ABC = 60^\circ$

b) Angle $PQR = 127^\circ$

c) Angle $XYZ = 275^\circ$

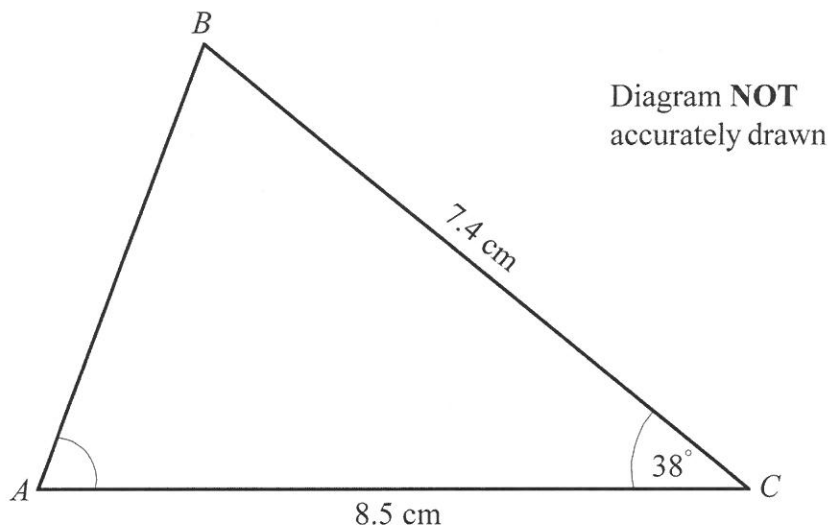
B A

P Q

X
 Y

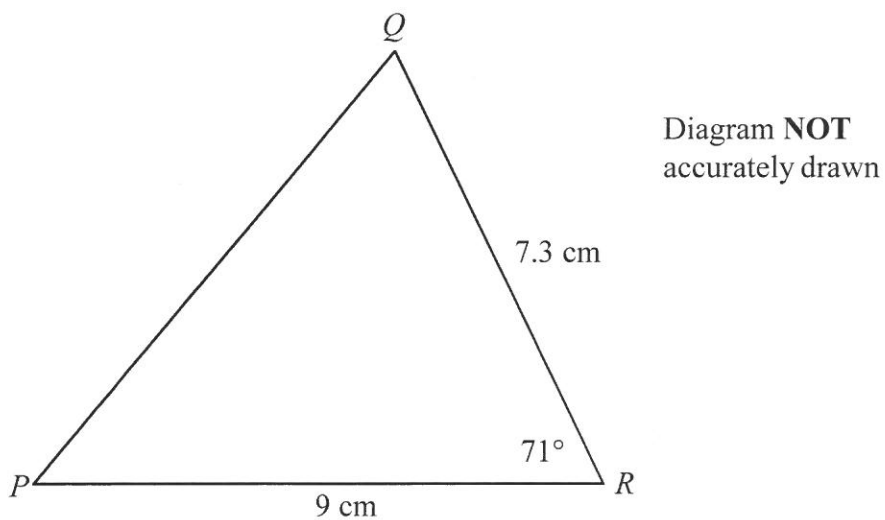
Drawing a Triangle Using a Protractor

- 1) The diagram shows a sketch of triangle ABC .



$BC = 7.4\text{ cm}$
 $AC = 8.5\text{ cm}$
 Angle $C = 38^\circ$

- Make an accurate drawing of triangle ABC .
 - Measure the size of angle A on your diagram.
- 2) The diagram shows a sketch of triangle PQR .

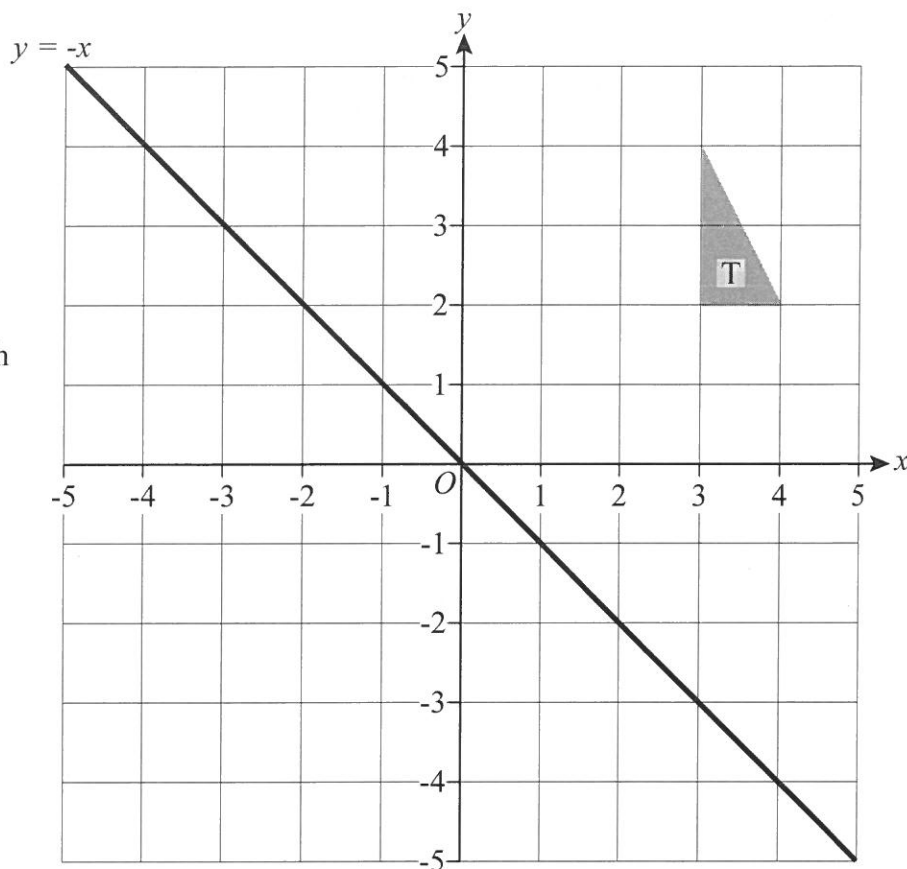


Make an accurate drawing of triangle PQR .

Reflections

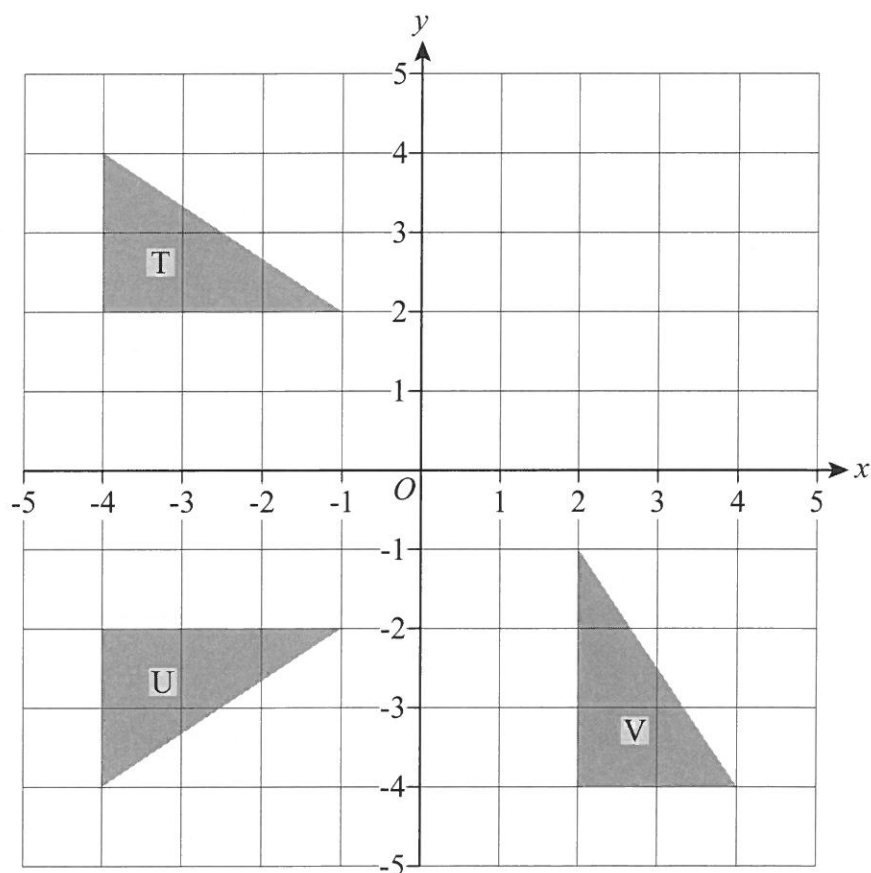
- 1) a) Reflect triangle T in the x axis.
Label your new triangle U.

- b) Reflect triangle T in the line with
equation $y = -x$.
Label your new triangle V.



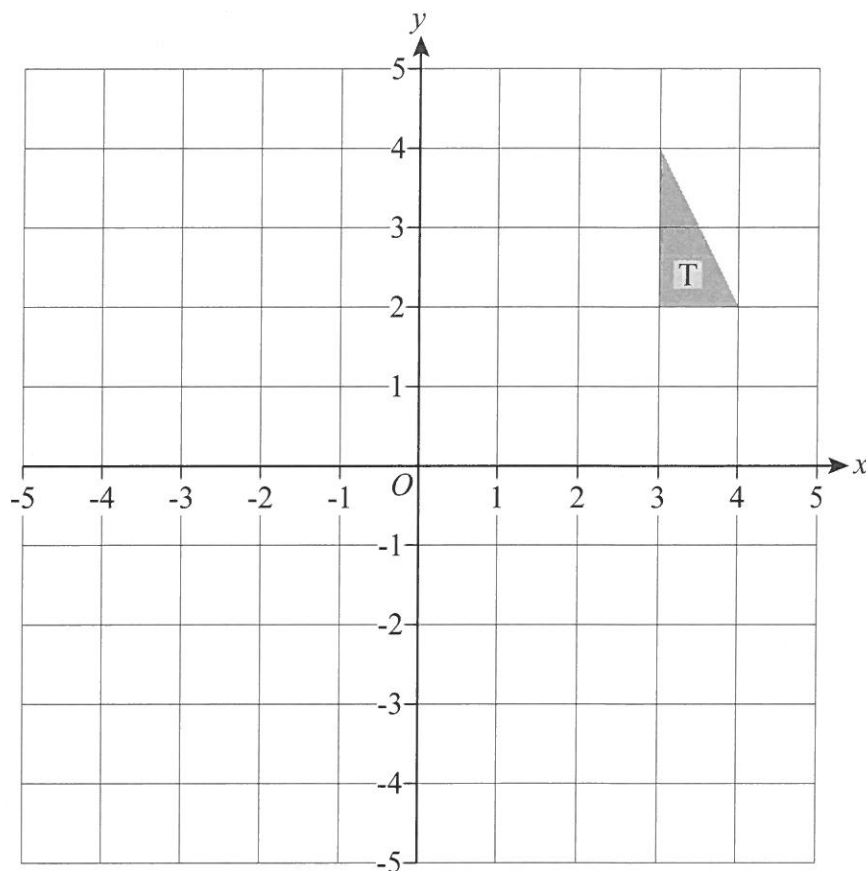
- 2) a) Describe fully the single
transformation which maps
triangle T to triangle U.

- b) Describe fully the single
transformation which maps
triangle T to triangle V.

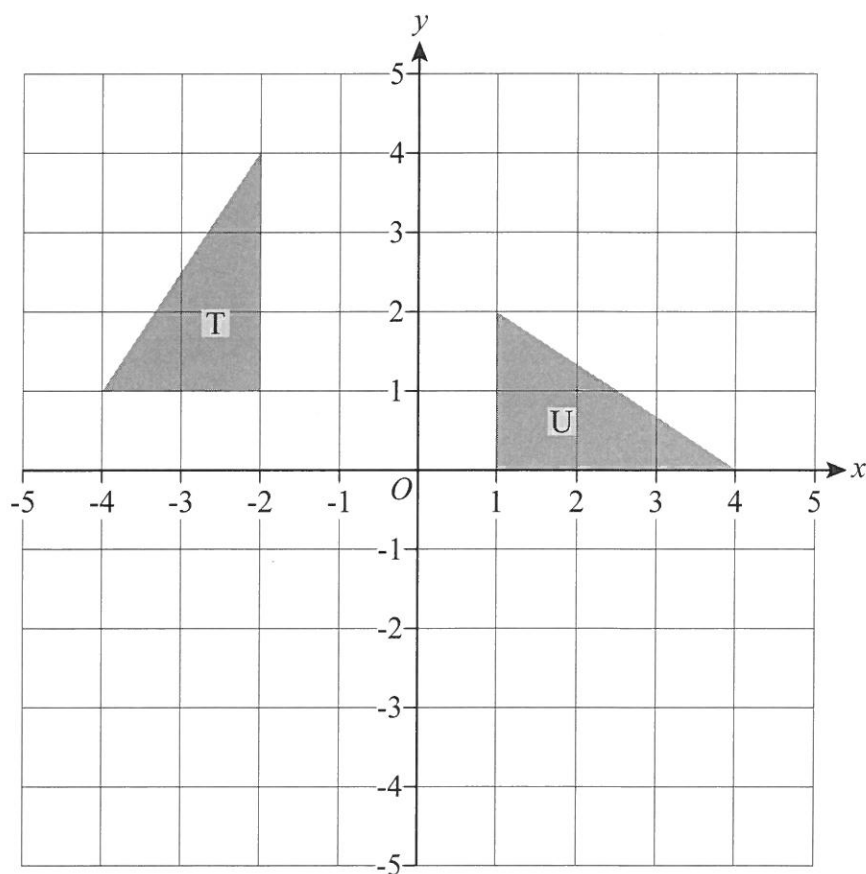


Rotations

- 1) a) Rotate triangle T 90° anti-clockwise about the point (0, 0).
Label your new triangle U.
- b) Rotate triangle T 180° about the point (2, 0).
Label your new triangle V.



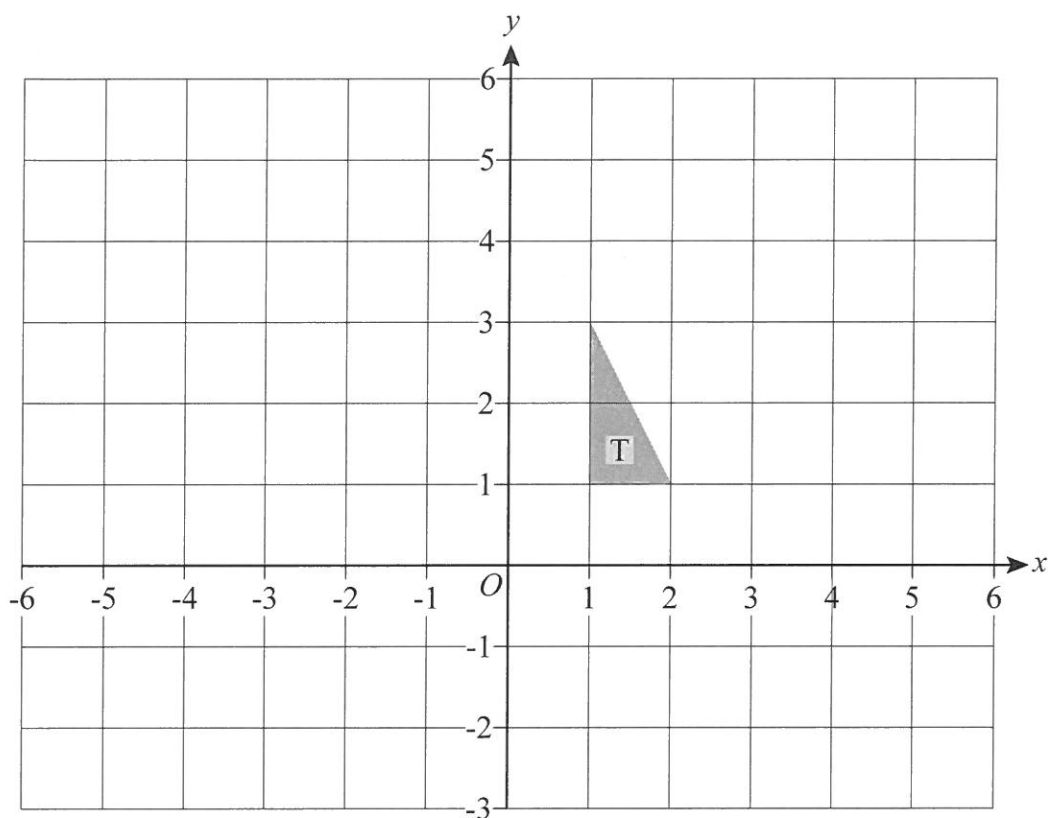
- 2) Describe fully the single transformation which maps triangle T to triangle U.



Translations

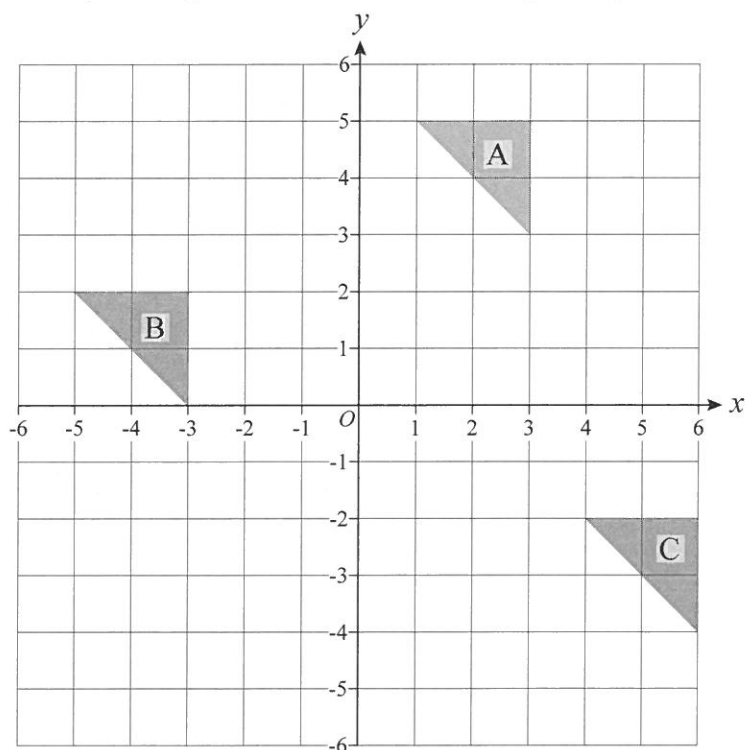
1) a) Translate triangle T by vector $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$ and label it U.

b) Translate triangle T by vector $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and label it V.

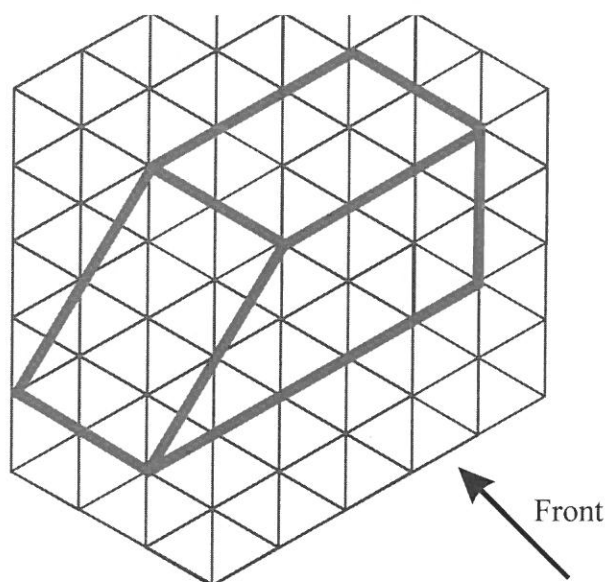


2) a) Describe fully the single transformation which maps triangle A to triangle B.

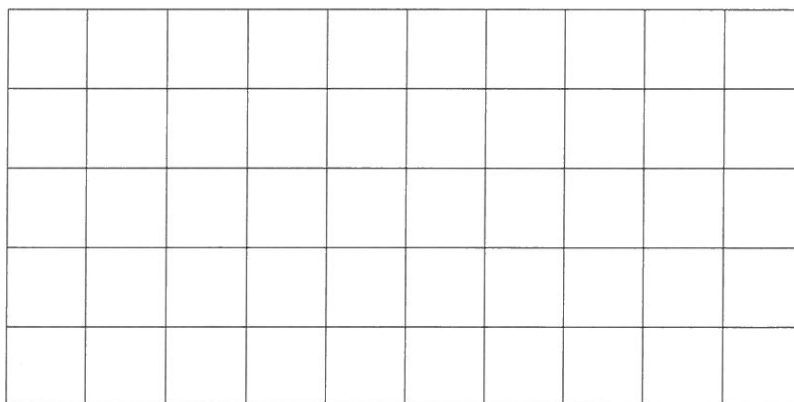
b) Describe fully the single transformation which maps triangle A to triangle C.



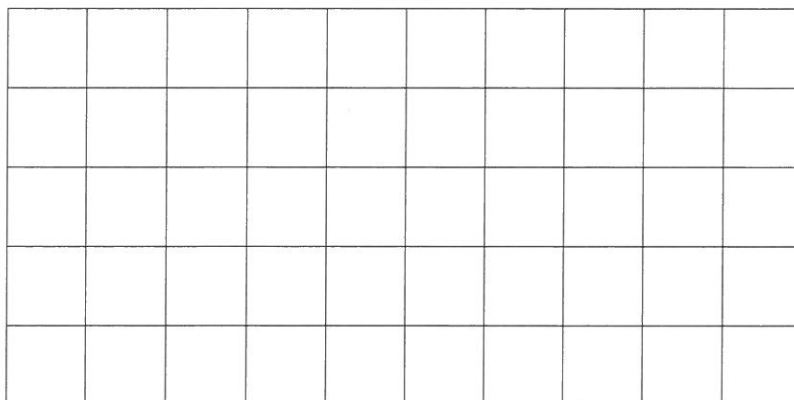
- 1) The diagram shows a prism drawn on an isometric grid.



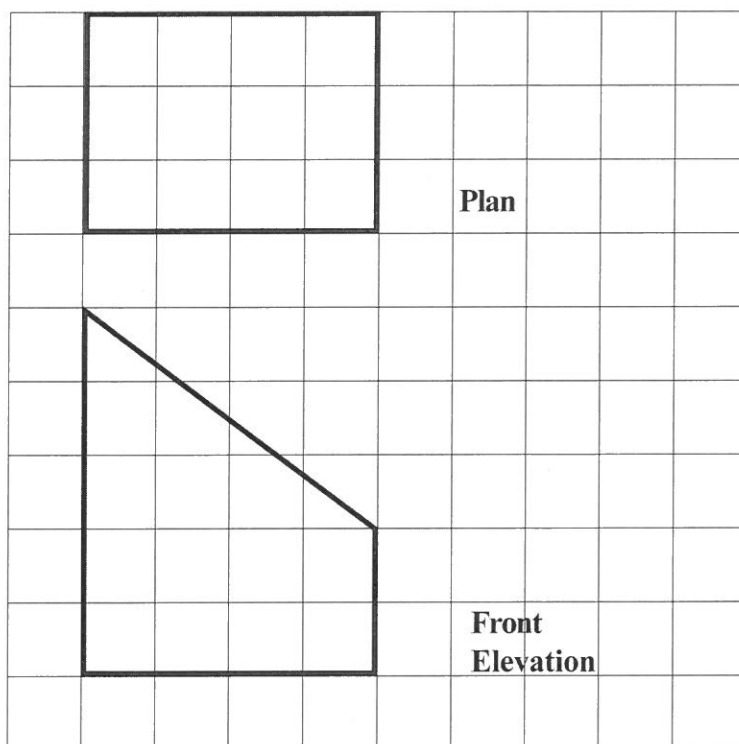
- a) On the grid below, draw the front elevation of the prism from the direction marked by the arrow.



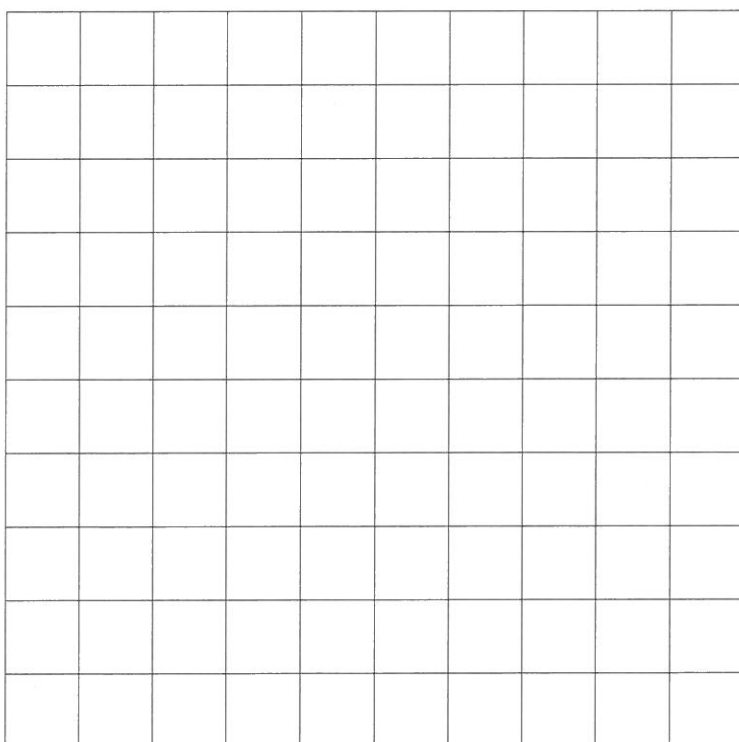
- b) On the grid below draw a plan of the prism.



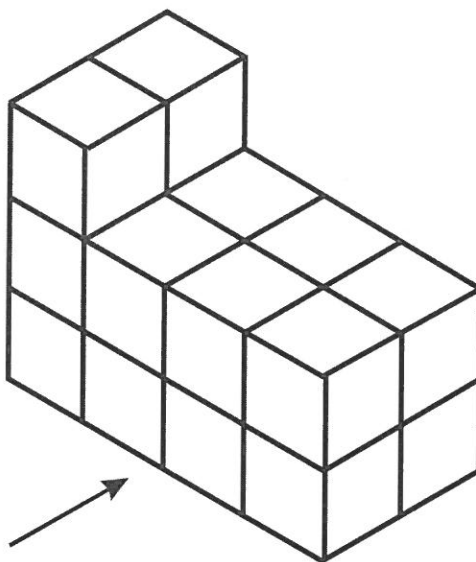
- 1) Here is the plan and front elevation of a prism.
The front elevation shows the cross section of the prism.



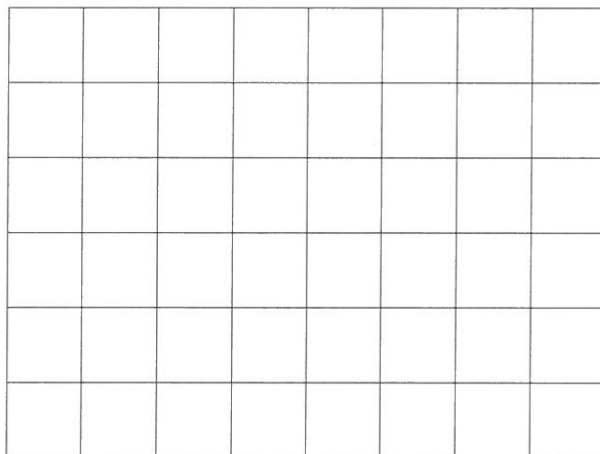
On the grid below, draw the side elevation of the prism.



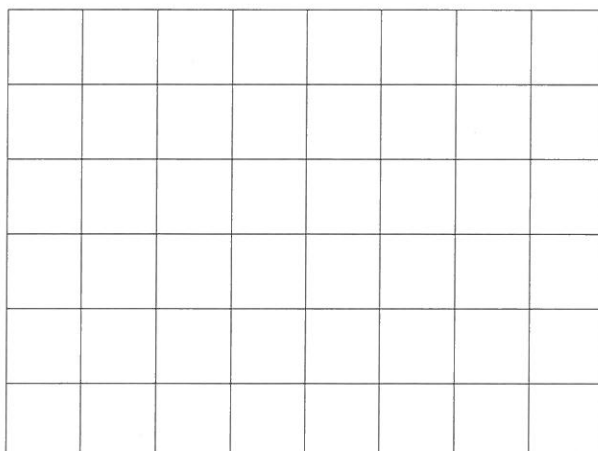
- 1) The diagram shows a solid prism made from centimetre cubes.



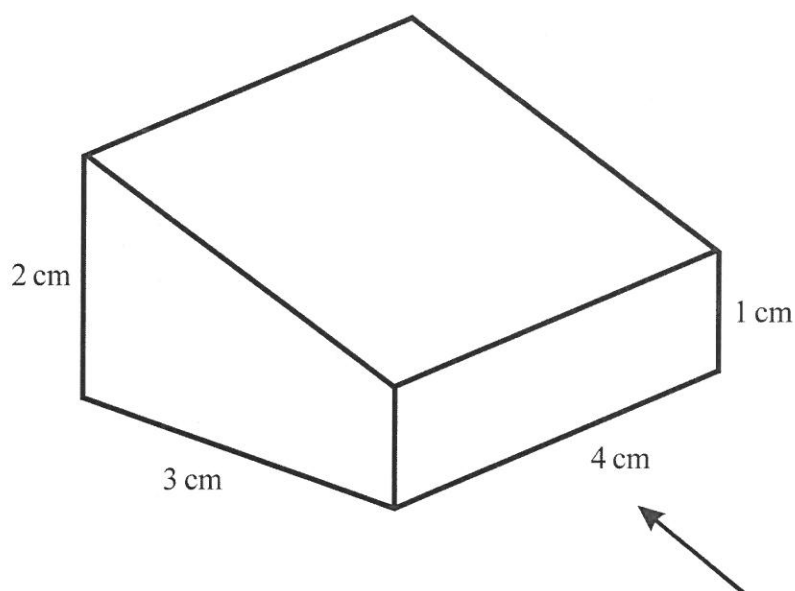
- a) On the centimetre square grid, draw the front elevation of the solid prism from the direction shown by the arrow.



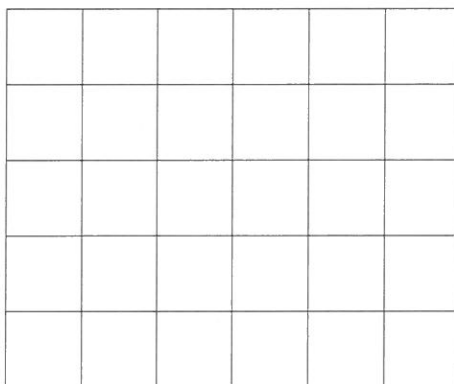
- b) On the centimetre square grid below, draw the plan of the solid prism.



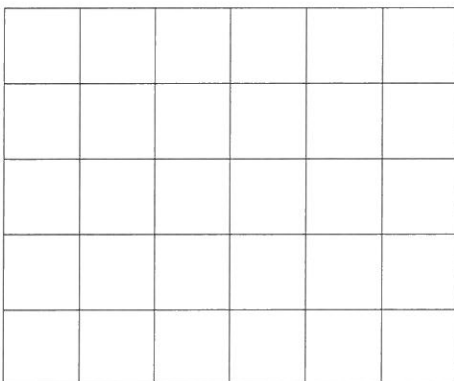
- 1) The diagram shows a solid prism.



- a) On the grid below, draw the front elevation of the prism from the direction of the arrow.

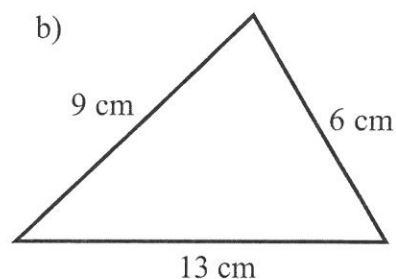
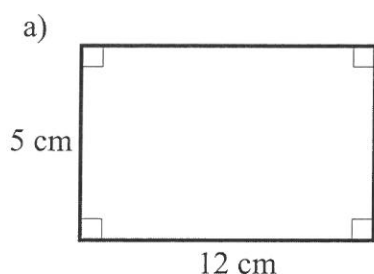


- b) On the grid below, draw the plan of the prism.



Perimeters

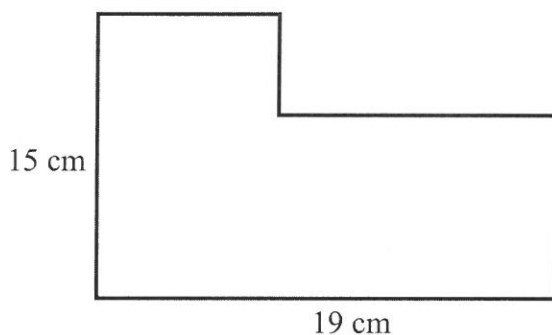
- 1) Find the perimeters of the following two shapes.



- 2) The length of a rectangle is 9 cm.
The total perimeter is 30 cm.

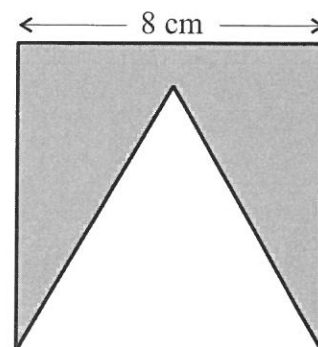
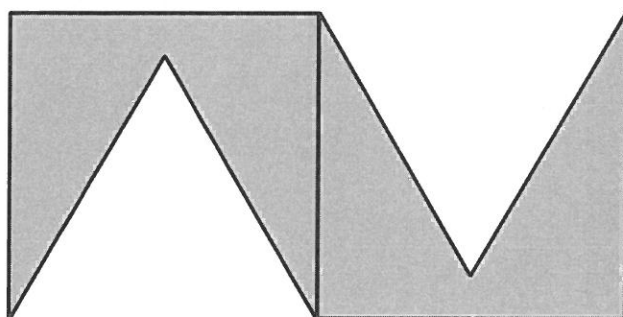
Calculate the length of the width of the rectangle.

- 3) Work out the perimeter of this L shape.



- 4) This shape is made by cutting out an equilateral triangle from a square.

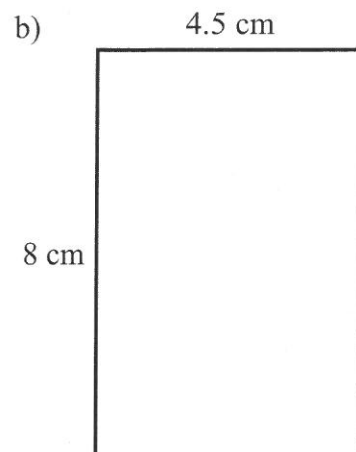
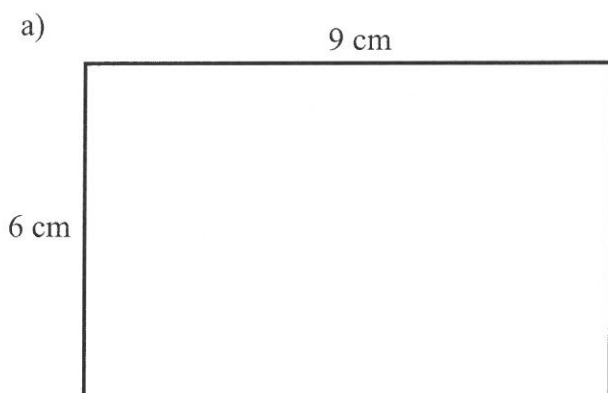
Two of these shapes are then put together to make this shape.



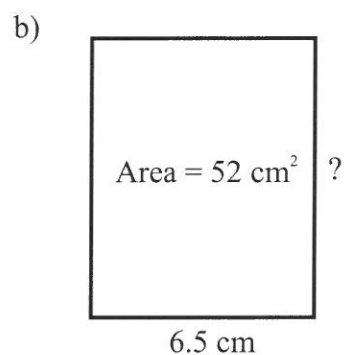
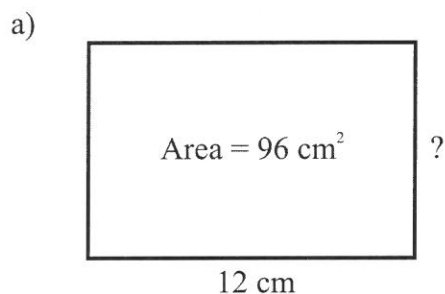
Work out the perimeter of this new shape.

Area of a Rectangle

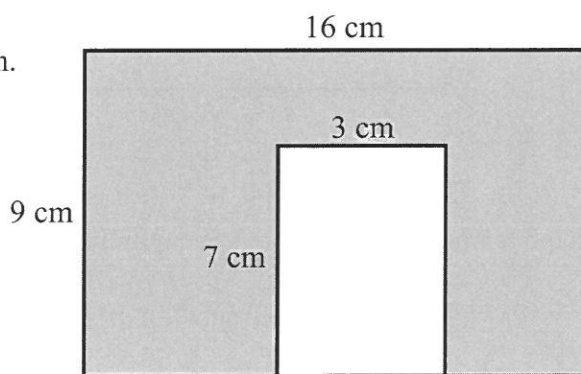
- 1) Find the areas of these two rectangles.



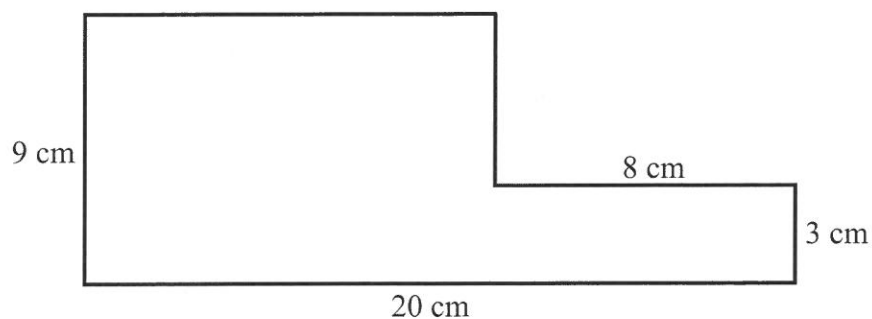
- 2) Find the size of the missing sides in these two rectangles.



- 3) Find the area of the shaded section.

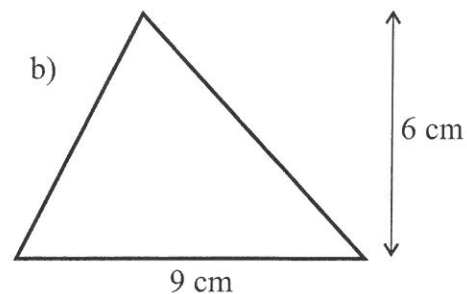
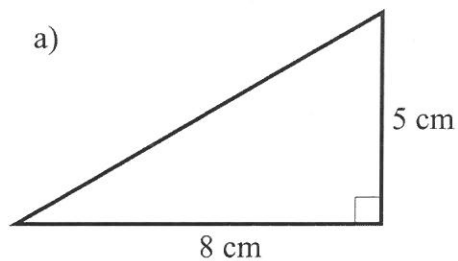


- 4) Find the area of the L shape.

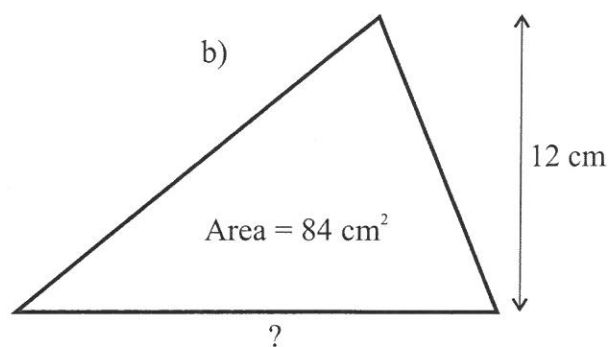
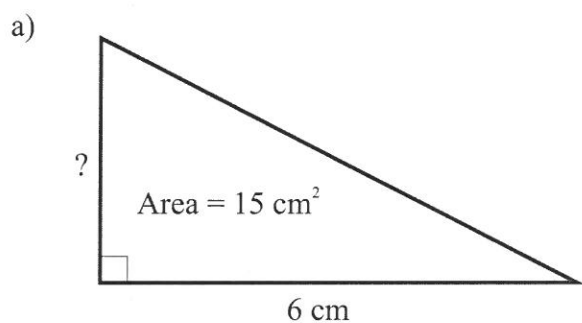


Area of a Triangle

- 1) Find the areas of the following two triangles.

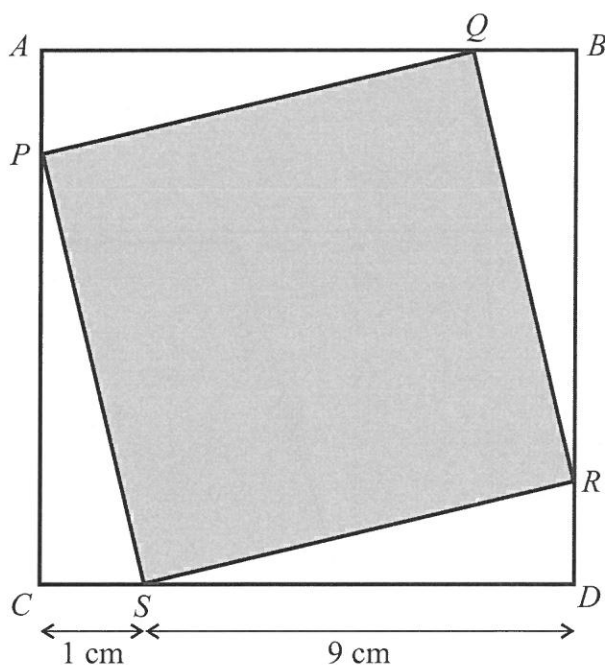


- 2) Find the missing lengths.



- 3) $ABCD$ is a square.

$PQRS$ is a square.

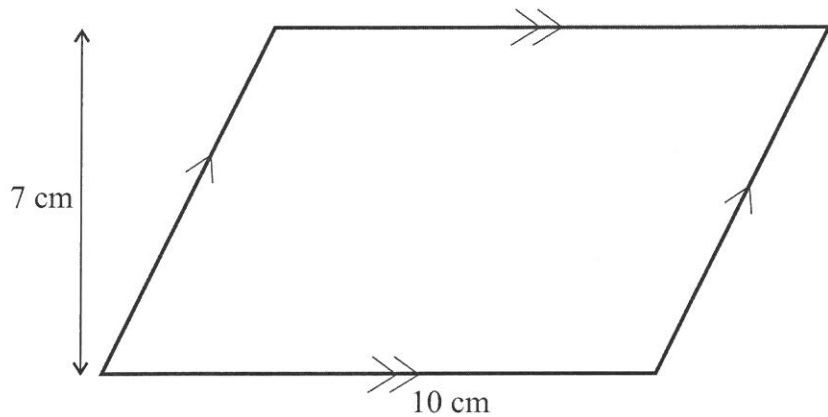


Find the area of the shaded square, $PQRS$.

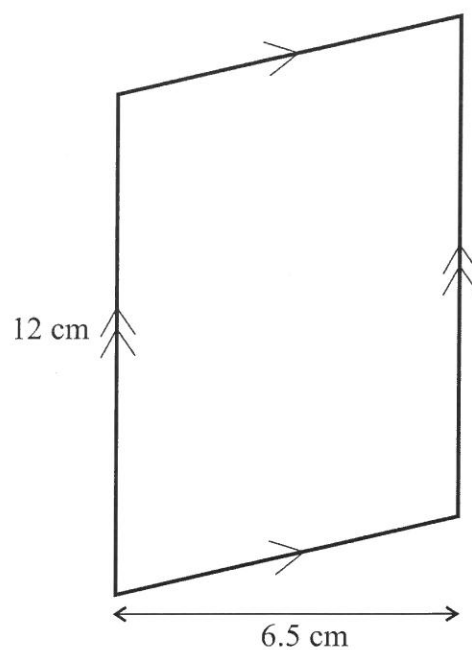
Area of a Parallelogram

- 1) Find the area of each of these parallelograms.

a)

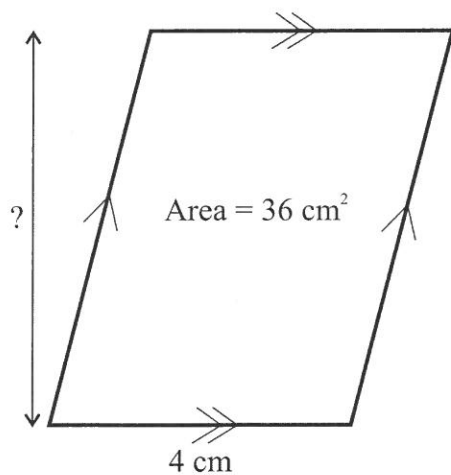


b)

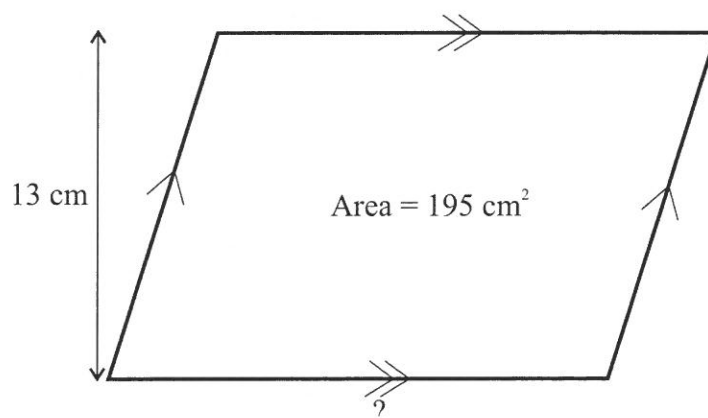


- 2) Find the missing lengths in these two parallelograms.

a)

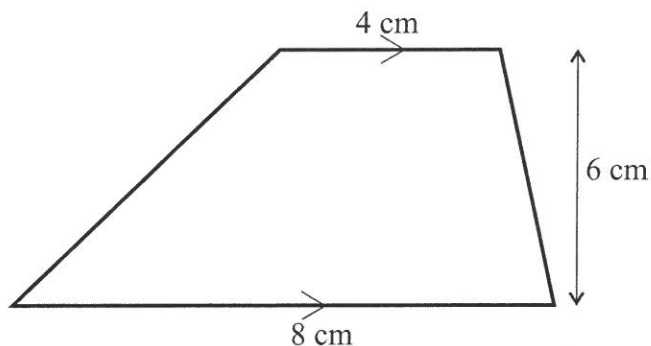


b)

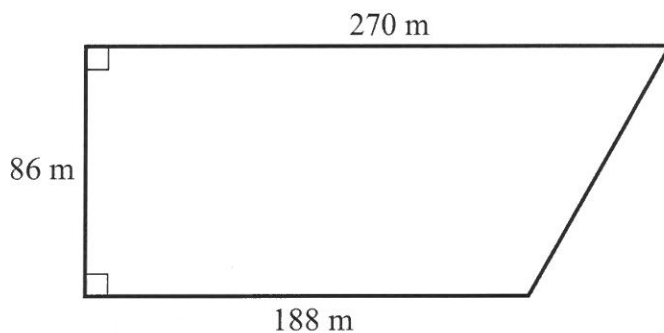


Area of a Trapezium

- 1) Find the area of this trapezium.



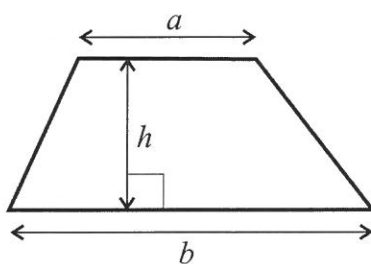
- 2) The diagram shows a field.



Work out the area of the field.



- 3) In the trapezium, $a = 6.6$ cm, $b = 8.4$ cm and $h = 3.6$ cm.

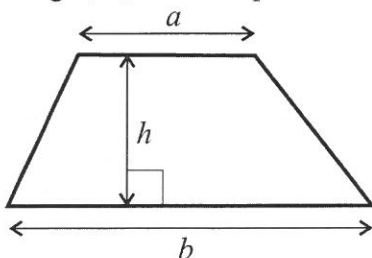


Work out the area of the trapezium.



- 4) In the trapezium below, the area is 45 cm^2 .
 $a = 5$ cm and $b = 10$ cm.

Calculate the height, h , of the trapezium.



Frequency Trees

100 people underwent an operation at a hospital.

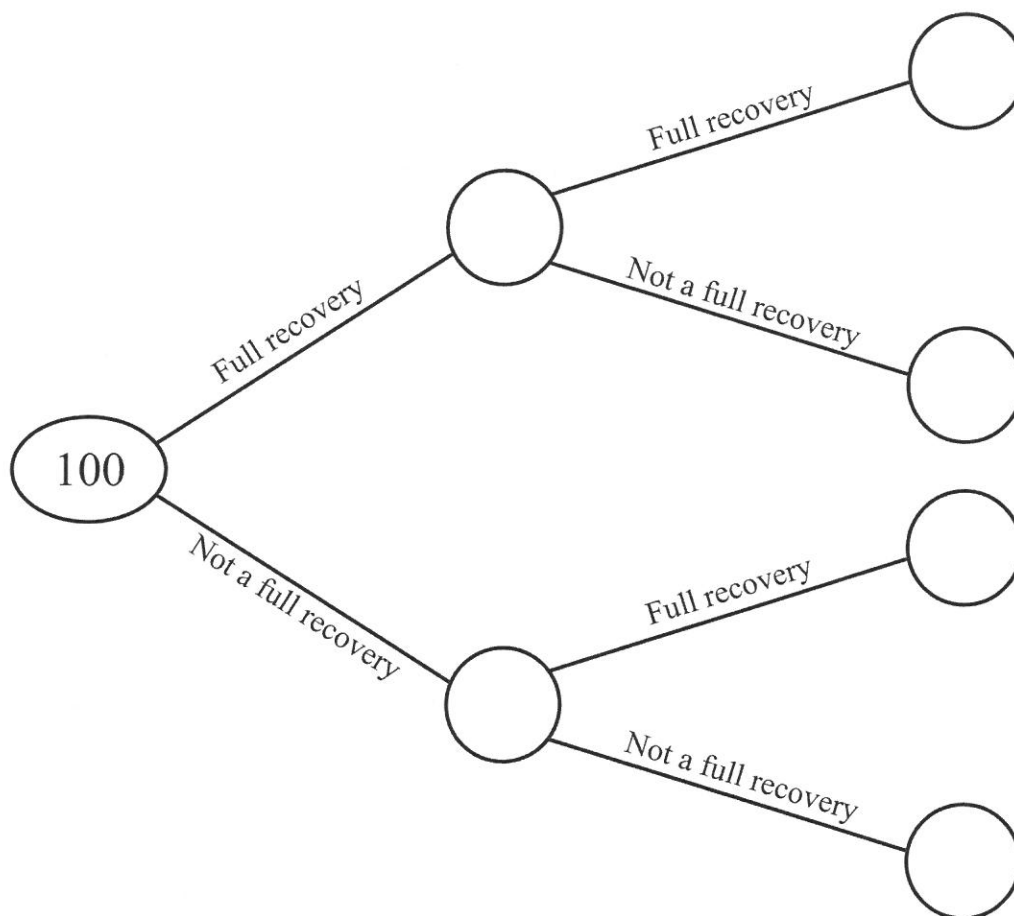
Before the operation, based on their medical notes, it was predicted whether each person would make a full recovery or not.

It was predicted that 85 people would make a full recovery.

It was later found that 6 of the people expected to fully recover, didn't.

Altogether, 87 people made a full recovery.

Complete the frequency tree.



- 1) List all the outcomes if two coins are flipped.

- 2)
 - a) How many possible outcomes are there if three coins are flipped?
 - b) List them all - the first one has been done for you: H H H

- 3)
 - a) How many possible outcomes are there if two six-sided dice are rolled?
 - b) List them all.

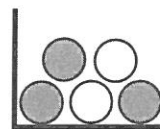
- 4)
 - a) How many possible outcomes are there if a coin is flipped and a dice is rolled?
 - b) List them all.

- 5)
 - a) How many possible outcomes are there if two coins are flipped and a dice is rolled?
 - b) List them all.

- 6) How many possible outcomes are there if 6 coins are flipped?

-  7) If Carly has each fingernail painted at a salon and can choose between red, blue and green for each nail, how many different combinations are there for her to choose from?

- 1) A box contains 3 grey counters and 2 white counters.
A counter is taken from the box at random.
What is the probability of choosing a white counter?



- 2) There are 3 blue counters, 5 red counters and 7 green counters in a bag.
A counter is taken from the bag at random.
- What is the probability that a green counter will be chosen?
 - What is the probability that a blue or red counter will be chosen?
- 3) In a class there are 10 boys and 15 girls.
A teacher chooses a student at random from the class.
Eric says that the probability a boy will be chosen is 0.5 because a student can be either a boy or a girl.
Jenny says that Eric is wrong.
Decide who is correct - Eric or Jenny - giving reasons for your answer.
- 4) Spinner A has numbers 1 to 4 on it.
Spinner B has numbers 1 to 3 on it.
Both spinners are spun and the numbers on each are added together to give a score.
What is the probability that the score will be
- 7?
 - 3 or 4?

Mutually Exclusive Events

- 1) If the probability of passing a driving test is 0.54, what is the probability of failing it?
- 2) The probability that a football team will win their next game is $\frac{2}{11}$.
The probability they will lose is $\frac{3}{11}$.
What is the probability the game will be a draw?



- 3) On the school dinner menu there is only ever one of four options.
Some of the options are more likely to be on the menu than others.
The table shows the options available on any day, together with three of the probabilities.

Food	Curry	Sausages	Fish	Casserole
Probability	0.36	0.41		0.09

- a) Work out the probability of the dinner option being Fish.
- b) Which option is most likely?
- c) Work out the probability that it is a Curry or Sausages on any particular day.
- d) Work out the probability that it is **not** Casserole.



- 4) Julie buys a book every week.
Her favourite types are Novel, Drama, Biography and Romance.
The table shows the probability that Julie chooses a particular type of book.

Type of book	Novel	Drama	Biography	Romance
Probability	0.24	0.16	x	x

- a) Work out the probability that she will choose a Novel or a Drama.
- b) Work out the probability that she will choose a Biography or a Romance.

The probability that she will choose a Biography is the same as the probability she will choose a Romance.

- c) Work out the probability that she will choose a Biography.

Two-Way Tables

- 1) Billy has been carrying out a survey.
He asked 100 people the type of water they like to drink (still, sparkling or both).
Here are part of his results:

	Still	Sparkling	Both	Total
Male	26			53
Female		20	10	
Total			16	100

- a) Complete the two-way table.
- b) How many males were in the survey?
- c) How many females drink only still water?
- d) How many people drink only sparkling water?
- 2) 90 students each study one of three languages.
The two-way table shows some information about these students.

	French	German	Spanish	Total
Female				
Male		7		
Total	20	18		90

50 of the 90 students are male.
29 of the 50 male students study Spanish.

- a) Complete the two-way table.
- b) How many females study French?
- c) How many people study Spanish?
- 3) Karen asks 100 students if they like milk, plain or white chocolates best.
36 of the students are girls.
19 of these girls like milk chocolates best.
16 boys like white chocolates best.
8 out of the 24 students who like plain chocolates best are girls.
Work out the number of students who like milk chocolates the best.



- 1) Kaya made a list of his homework marks.

3 2 3 4 1 4 5 4

- Write down the mode of Kaya's marks.
- Work out his mean homework mark.

- 2) Lydia rolled an 8-sided dice ten times.
Here are her scores.

5 1 2 5 3 8 6 6 3 2

- Work out Lydia's median score.
- Work out the mean of her scores.



- 3) In a two-week period, a train was this many minutes late each day:

3 0 0 0 7 4 5 2 0 1 14 0 5 1

- What was the mean average number of minutes late?
- What was the median average number of minutes late?



- 4) Two small Year 10 classes, Set A and Set B, sat the same Science test.

Set A had these scores for the test:

63%, 71%, 48%, 95%, 46%, 82%, 77%, 36%, 73%

Set B had these scores:

58%, 63%, 85%, 61%, 59%, 38%, 90%, 84%, 75%, 48%

How much bigger was Set B's mean average score than Set A's mean average score?
Give your answer correct to 1 decimal place.



- 5) A rugby team played six games.

The mean score for the six games is 15

The rugby team played one more game.

The mean score for all seven games is 16

Work out the number of points the team scored in the seventh game.

- 1) Out of the following types of data, decide which is continuous and which is discrete:

The lengths of some roads.

The number of 'cats eyes' on a one mile stretch of road.

The time it takes twenty students to complete an English essay.

The number of pages in twenty students English essays.

The weights of sacks of potatoes.

The number of potatoes in some sacks of potatoes.

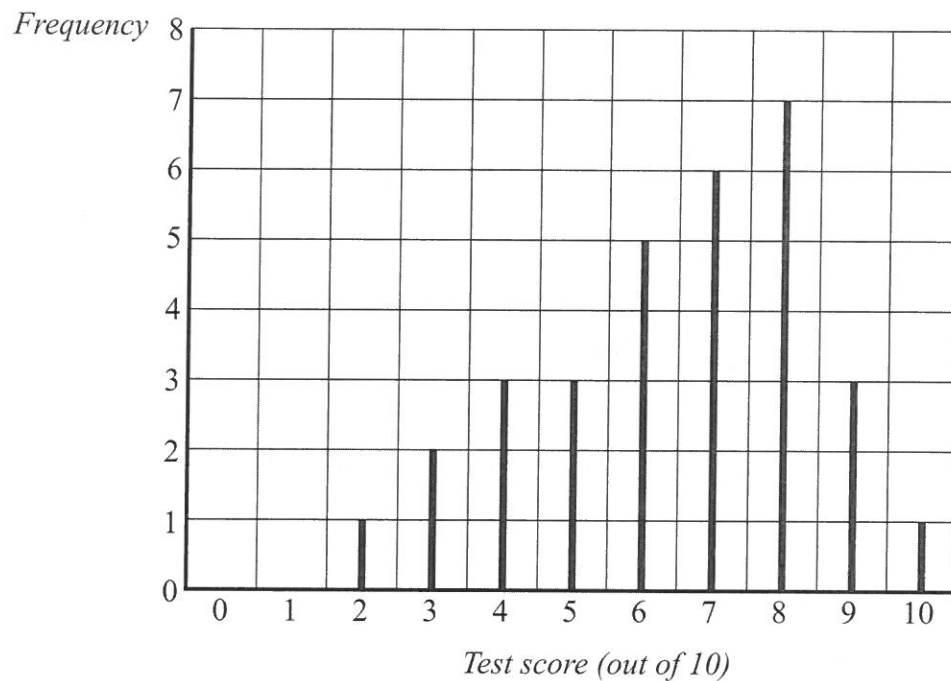
The depth of water as the tide comes in and goes out.

The number of crackers in some packets of biscuits.

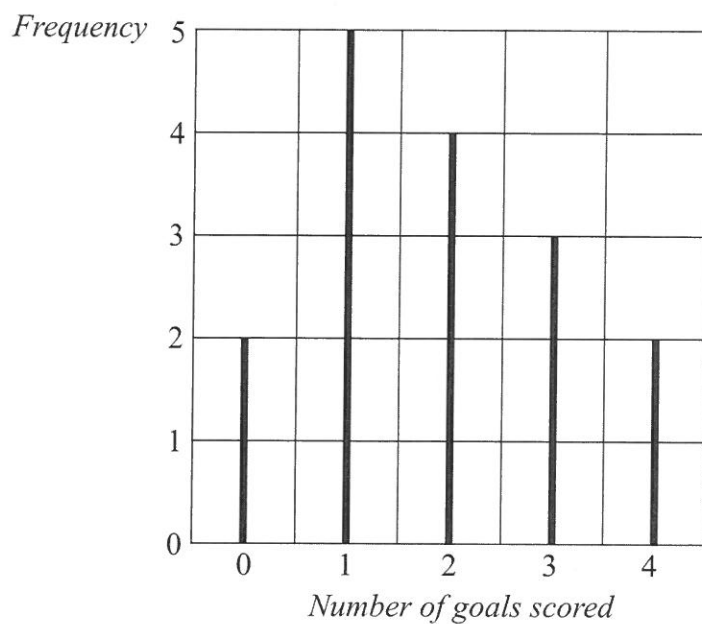
The weight of the crackers in some packs of biscuits.

- 2) Write a short statement which explains what continuous data is.
- 3) Write a short statement which explains what discrete data is.

- 1) The line chart shows the test scores of pupils in class 10A.



- How many pupils sat the test in class 10A?
 - What was the modal test score?
 - What was the median test score?
- 2) The line chart shows the number of goals scored by a football team in their last 16 games.



- How many goals did they score altogether?
- What was the median number of goals scored?

Frequency Tables and Diagrams

- 1) A class of pupils is asked to solve a puzzle.
The frequency table below shows the times taken by the pupils to solve the puzzle.

Time (t) in min	Frequency
$0 < t \leq 5$	3
$5 < t \leq 10$	4
$10 < t \leq 15$	5
$15 < t \leq 20$	7
$20 < t \leq 25$	5

Draw a frequency diagram to show this information.
